

Siddhartha Kumar Senthil Kumar

Data Analyst | Software Engineer | Machine Learning

121 Centennial Court, Burlington, VT 05401 | 201-377-2170 | siddhartha.kumar.sk.1998@gmail.com
github.com/siddharthakumar-98 | siddharthakumar-98.github.io

PROFESSIONAL SUMMARY

Data Analyst and prior Software Engineer with **4+ years** of experience delivering end-to-end analytics solutions across healthcare and finance. Strong background in statistical analysis, predictive modeling, and Python/SQL engineering, with applied experience in machine learning, computer vision, vision-language models, and AI-assisted workflows. Skilled at building cloud data pipelines, self-service BI dashboards, and production full-stack applications that translate complex data into measurable business outcomes.

TECHNICAL SKILLS

Languages	Python, SQL, R, C, C++, Rust, Java, JavaScript, TypeScript, Node.js
ML & AI	PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, OpenCV, YOLO, MLX; LLMs/VLMs, Computer Vision, NLP, RAG, Fine-tuning, Reinforcement Learning
Data & Statistics	Pandas, NumPy, SciPy, statsmodels; EDA, Hypothesis Testing, A/B Testing, ANOVA, Cohort & Funnel Analysis, Time Series, Predictive Modeling
Visualization & BI	Tableau, Power BI, Looker, Matplotlib, Seaborn, Advanced Excel (Pivot Tables, Power Query, VBA)
Data Platforms & Cloud	AWS (S3, Glue, Redshift, Lambda, Athena, Bedrock), GCP, Snowflake, Databricks, Spark, Kafka, Airflow, dbt, SSIS
Databases & Web	PostgreSQL, MySQL, Oracle, MongoDB, Supabase; Next.js, React, Tailwind CSS, Vercel
Tools & Practices	Git, Docker, Kubernetes, CI/CD, Ghidra, JIRA; Agile/Scrum, SDLC, ETL/ELT; HIPAA, SOX, GDPR

PROFESSIONAL EXPERIENCE

Data Analyst | **Optum** | Remote, USA Dec 2024 – Present

- Conducted large-scale **EDA** on **95M+ Medicare claims and EHR records** using ML-assisted feature extraction, uncovering readmission drivers and seasonal trends and improving predictive analytical accuracy by **41%**.
- Applied **ML classification, regression, and hypothesis testing** to evaluate care coordination, identifying key readmission predictors via feature engineering and contributing to a **19% reduction** in 30-day readmission rates.
- Applied predictive modeling and clustering to patient demographics, diagnoses, and utilization, identifying high-risk cohorts and improving early readmission risk detection by **26%**.
- Automated **Python ETL** with AI-assisted validation for Medicare datasets, reducing manual processing by **58%** and cutting daily analytics workflows from 5 hours to 2.
- Built **Tableau dashboards** surfacing AI-generated risk stratification scores, readmission KPIs, and care gap insights, boosting executive decision-making speed by **38%** and driving self-service analytics adoption.
- Led **Agile Scrum** delivery across biweekly sprints with clinicians, care managers, and product owners, cutting readmission reporting delays by **32%**.
- Engineered **SSIS pipelines** integrating 18+ sources into an **AWS S3** data lake for ML serving, improving refresh reliability by **72%**, reducing latency by **44%**, and lowering infrastructure costs by **27%**.

Software Engineer II | **JPMorgan Chase** | Dallas, TX Dec 2020 – Dec 2022

- Delivered 5 enterprise analytics releases on schedule across all **SDLC phases**, improving reporting accuracy by **35%**.
- Performed **hypothesis-driven analysis** with statistical modeling and anomaly detection on revenue, expense, and variance data, reducing reporting discrepancies by **22%** annually.
- Cleaned and standardized **12M+ financial records**, improving dataset reliability by **46%**; optimized complex **SQL** (CTEs, joins, window functions, query plans), cutting execution time by **36%**.
- Optimized **MySQL** schema design and indexing for enterprise financial datasets, improving retrieval speed by **31%**.
- Designed **Power BI dashboards** for revenue, cost-driver, and profitability KPIs, improving executive visibility by **40%**.
- Enforced **SOX-compliant data governance** via access controls, validation checks, and audit trails; applied SAS and Python for variance modeling, improving forecasting accuracy by **28%**.

RESEARCH EXPERIENCE

Graduate Teaching & Research Assistant | University of Vermont

Jan 2023 – Dec 2024

- Developed ML assignments in **Computer Vision, Deep Learning, and Neural Networks** (Dr. Safwan Wshah); built synthetic datasets with **bias mitigation** for Inclusive Computing (Prof. Yuanyuan Feng).
- Conducted **cognitive computing research** detecting mind-wandering from physiological signals with ML classification (Prof. David Jangraw); trained **reinforcement learning** robotic agents in PyBullet/Taichi.

EDUCATION

M.S. in Computer Science | University of Vermont | Burlington, VT

Jan 2023 – Dec 2024

B.Sc. in Computer Engineering | Purdue University | West Lafayette, IN

Aug 2016 – May 2021

PROJECTS

QuantiPhy | NeurIPS 2026 Competition Track: Quantitative Physical Reasoning from Video

Sep 2026 – Present

Stack: Python, PyTorch, YOLO11-seg, ByteTrack, OpenCV, Qwen3-VL / Qwen3.5 (4-bit), Apple MLX, CUDA (Colab), Meta VGGT, pytest

- Built an end-to-end system that answers numerical physics questions from video, estimating **size, speed, acceleration, and distance** for **3,289 questions across 568 videos**, scored by mean relative accuracy (MRA) over four 2D/3D, static/dynamic categories.
- Designed a **hybrid computer-vision + VLM architecture**: YOLO segmentation with dense ByteTrack tracking, metric calibration from supplied physical priors, and robust Huber-weighted polynomial kinematics, with **4-bit Qwen VLMs** as fallback whenever geometry abstains.
- Improved public-validation MRA from **37.5% to 46.0% (+8.6 pp)** through 8 controlled, logged experiments: automatic foreground tracking, camera-range consistency checks, and diameter-calibrated circle tracking with HSV color grounding (2D-static category **26.9% → 50.6%**).
- Kept only what the data supported: rejected a larger **9B model** and a model-consensus approach after both regressed, and routed questions using only question text, priors, and video, never validation labels.
- First official submission scored **36.8% MRA on the hidden test set**; traced its **10.7% invalid predictions** to zero outputs through error analysis and built a targeted VLM retry pass to recover them.
- Ran quantized inference locally on an **M4 Pro with MLX** and at scale on **Colab GPUs** with resumable checkpoints; added SHA-256-verified data downloads, provenance hashes, a strict submission exporter, and **100+ unit tests**.

GameMerge | PS2 Reverse Engineering & Cross-Game Gameplay Mod

2026 – Present

Stack: Rust, C, MIPS R5900 assembly, Ghidra, PCSX2 PINE IPC, ps2dev toolchain, Docker

- Architecting a **passthrough mod** that runs *Burnout 3: Takedown's* gameplay (takedowns, boost, slow-mo, Crash mode) inside *Midnight Club 3*: two PCSX2 emulators linked by a **Rust bridge** that syncs per-frame shared-memory mailboxes over PINE.
- Leading a **byte-matching decompilation** of Burnout 3 (9,107 functions): reproduced the original executable **byte-for-byte (SHA-1 match)**, identified the original CodeWarrior 3.0.3 compiler, mapped the binary's memory layout, and named **260 library functions** in Ghidra.
- Wrote a **Ghidra export-repair tool** for a PS2 static recompiler that split mis-sized functions at data boundaries, cutting **4,270 unhandled instruction encodings to zero** and generating 71K+ C++ files.
- Planned a staged migration that compiles decompiled gameplay rules into the host mod one system at a time, validating each against the original game until outputs agree.

LetterLane | Real-Time Multiplayer Word Game **Monetization-ready**

letterlane.vercel.app | 2026

Stack: Next.js, React, TypeScript, Tailwind CSS, Supabase (Auth, PostgreSQL, Realtime), Zod, Vitest, Playwright, Vercel

- Built and deployed a full-stack, real-time **two-player word and phrase game** (duel and co-op, three bot difficulty levels) with a server-authoritative engine, row-level security, and row locking plus idempotency keys to protect game state.
- Prepared the game for revenue in v3.0 with **Google AdSense and H5 Games ad placements**, consent and privacy controls, and SEO pages, giving it a clear path to monetization as an ad-supported web game.
- Built bot opponents using **constraint-based candidate filtering** and information-ranked word selection; tested with **Vitest, Playwright E2E** (desktop and mobile), and axe-core accessibility checks.